



Cutting-edge therapeutic ultrasound-its basic and clinical medicine; the spread of ultrasound-based theranostics

Ultrasound and microbubble-mediated drug delivery and immunotherapy

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Abstract

Ultrasound induces the oscillation and collapse of microbubbles such as those of an ultrasound contrast agent, where these behaviors generate mechanical and thermal effects on cells and tissues. These, in turn, induce biological responses in cells and tissues, such as cellular signaling, endocytosis, or cell death. These physiological effects have been used for therapeutic purposes. Most pharmaceutical agents need to pass through the blood vessel walls and reach the parenchyma cells to produce therapeutic effects in drug delivery. Therefore, the blood vessel walls act as an obstacle to drug delivery. The combination of ultrasound and microbubbles is a promising strategy to enhance vascular permeability, improving drug transport from blood to tissues. This combination has also been applied to gene and protein delivery, such as cytokines and antigens for immunotherapy. Immunotherapy, in particular, is an attractive technique for cancer treatment as it induces a cancer cell-specific response. However, sufficient anti-tumor effects have not been achieved with the conventional cancer immunotherapy. Recently, new therapies based on immunomodulation with immune checkpoint inhibitors have been reported. Immunomodulation can be regarded as a new strategy for cancer immunotherapy. It was also reported that mechanical and thermal effects induced by the combination of ultrasound and microbubbles could suppress tumor growth by promoting the cancer-immunity cycle via immunomodulation in the tumor microenvironment. In this review, we provide an overview of the application of ultrasound and microbubble combination for drug delivery and activation of the immune system in the microenvironment of tumor tissue.

Keywords Microbubbles · Drug delivery system · Blood–brain barrier opening · Immunomodulation · Cancer immunotherapy

Introduction

Ultrasound is used for diagnosis and therapy in the clinic. Ultrasound imaging is used in the detection of blood flow and tissue lesions, whereas ultrasound ablation is utilized in the treatment of essential tremor and cancer. Microbubbles as contrast agents are used to acquire enhanced images in ultrasound-based diagnoses. The mechanical and thermal effects induced by the combination of ultrasound and microbubbles have been applied to enhance vascular and cellular membrane permeability. This has been studied extensively for therapeutic applications, such as drug delivery and immunomodulation [1, 2]. In drug delivery, a novel system is required to improve the therapeutic effects of drugs. In recent years, pharmaceutical agents have dramatically diversified to include chemical compounds with small

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molecular weights, larger antibodies, and drug-loaded nano-carriers. In addition, cell-based therapy has been approved to treat cancer. It is vital that most medicines pass through the blood vessel walls and reach the parenchyma tissues to exert their therapeutic effects. Therefore, several approaches to enhance their vascular permeability, such as using biologically active substances and physical energy, have been studied. The combination of ultrasound and microbubbles is a promising approach with noninvasiveness and selectivity to enhance the vascular permeability followed by drug delivery. In immunomodulation, especially in cancer therapy, an innovative system is required to maximize the therapeutic effect, such as immune checkpoint blockade and chimeric antigen receptor-T cells. Although these drugs are a breakthrough for cancer therapy, the overall response rate could be improved. To enhance the effects of these agents, changing the micro-environment of the tumor is essential. Enhanced delivery of drugs/chemicals, such as cytokines, via the combination of ultrasound and microbubbles is a promising strategy. Other therapeutic effects induced by this combination are also attractive for immunotherapy. In this review, we provide an overview of the benefits of combination of ultrasound and microbubbles for drug delivery and activation of the immune system in cancer therapy and describe the future prospects.

Drug delivery by the combination of microbubbles and ultrasound exposure

Most drugs need to pass through blood vessel walls and interact with their target molecules. Although the permeability of blood vessel walls is different in normal and diseased tissues and varies depending on the organs, blood vessel walls with low permeability are one of the major obstacles to drug delivery. Therefore, controlling vascular permeability is essential to achieve an efficient therapeutic response. Recently, the combination of ultrasound and microbubbles has been explored for therapeutic applications, and reports suggest that it can enhance the permeability of blood vessel walls, resulting in increased drug transfer into tissues. In particular, drug delivery systems using this combination that target tumors have received much attention.

The combination of ultrasound and microbubbles has been applied to tumor-targeted drug delivery to enhance vascular permeability [3]. However, this mechanism is not fully understood. The behavior of microbubbles induced by ultrasound exposure plays a critical role in enhancing vascular permeability [4, 5]. Ultrasound exposure induces oscillation (repeat of compression and expansion; non-inertial cavitation) [6, 7] and collapse (inertial cavitation) [8] of microbubbles depending on several parameters, such as the frequency, intensity, and duration of ultrasound and the properties of the microbubbles [9] (Fig. 1a). Microstreaming

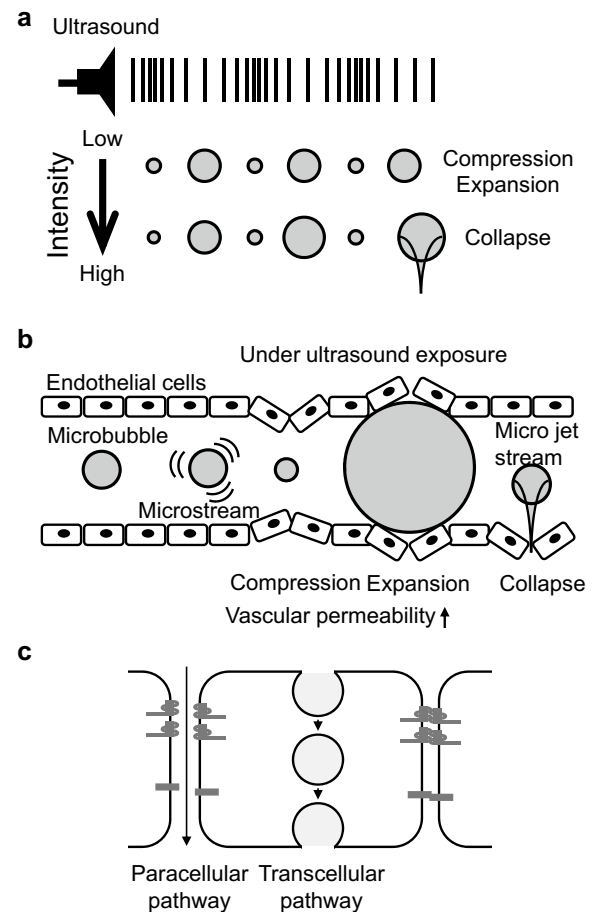


Fig. 1 Microbubble behavior and enhancement of vascular permeability with ultrasound and microbubbles. **a** The behavior of microbubbles induced by ultrasound. **b** Enhanced vascular permeability induced by ultrasound and microbubbles. **c** Drug delivery via paracellular and transcellular pathways

induced by oscillation results in the induction of shear stress in endothelial cells [7]. Compression pulls endothelial cells, while expansion pushes them [6]. Microjet streaming originating from the collapse of microbubbles results in the formation of transient pores on the endothelial membrane, and in various conditions, cell death and hemorrhage, i.e., the extravasation of red blood cells into the tissues [8]. These mechanical effects are involved in the enhancement of vascular permeability (Fig. 1b).

Brain tumor-targeted drug delivery

The blood–brain barrier (BBB), which is composed of endothelial cells and covered by pericytes and astrocytes, protects the brain by regulating the transport of molecules [10]. The endothelial cells interact with each other through tight junctions. Only compounds with low molecular weight and high hydrophobicity can pass through the BBB,

as against most drugs [11, 12]. In brain tumors, the blood vessels possess different characteristics to that of the BBB, and this is called the brain–tumor barrier (BTB) [13, 14]. Although the BTB is more permeable than the BBB, permeable vasculature heterogeneously exists, making the BTB an obstacle for chemotherapy. Through a combination of ultrasound and microbubbles, enhanced vascular permeability in the brain was achieved mainly via the paracellular and transcellular pathways (Fig. 1c) [15]. The detailed mechanism of enhanced vascular permeability with the combination of ultrasound and microbubbles has already been well summarized in the literature [16].

Brain tumor-targeted drug delivery using the combination of ultrasound and microbubbles has been extensively investigated because of the high level of malignancy of and lack of promising therapy for brain tumors. Anti-cancer drugs such as temozolomide [17], carboplatin [18, 19], doxorubicin [20], paclitaxel [21], bis-chloroethylnitrosourea [22], and etoposide [23] were efficiently delivered to tumor cells in the brain after treatment with the combination of ultrasound and microbubbles. In addition, this combination can efficiently deliver liposomal doxorubicin [24–26], liposomal paclitaxel [27], and cisplatin-loaded nanoparticles [28, 29] to brain tumors. Proteins, such as anti-CD47 antibody [30], bevacizumab [31], trastuzumab [32], and interleukin-12 [33], have also been delivered. Efforts were made to reduce resistance to anti-cancer drugs and improve efficacy where methyltransferase-targeted therapy was studied, and liposomal drugs [34] and plasmid DNA [35] were delivered to brain tumors by ultrasound and microbubble combination. Based on these promising studies, some clinical studies have been conducted using ultrasound devices.

An ultrasound platform for treating essential tremor and tremor-dominant Parkinson's disease called Exablate Neuro (InSightec) has 1024 elements arranged at a hemispherical transducer and transcranial 220-kHz focused ultrasound, and is performed under the guidance of magnetic resonance (MR) imaging. A clinical safety and feasibility study of this technique was reported in 2019 [36]. In this study, five patients with glioma were recruited; one patient was given liposomal doxorubicin and four patients received temozolomide before transcranial MR-guided focused ultrasound treatment. The sonicated position was defined by a cubic 3-by-3 grid with 3-mm spacing, approximately $9 \times 9 \times 6 \text{ mm}^3$. Definity, which is an ultrasound contrast agent, was used in this study. There were no adverse clinical or radiological events related to the procedure. In addition, enhancement of signal intensity derived from the MR contrast agent was observed in four patients after MR-guided focused ultrasound treatment, and an increase in the drug concentration in the sonicated area was achieved compared with that in the un-sonicated area in two patients. These results suggest that enhancing the vascular permeability in

tumors with ultrasound and microbubbles is feasible and does not induce notable side effects.

SonoCloud (CarThera) is composed of a 10-mm-diameter ultrasound transducer with a frequency of 1 MHz and is covered with a 11.5-mm-diameter housing. The transducer is implanted under the skull, and power is supplied outside. Clinical trials for safety and feasibility were reported in 2014 [37] and 2019 [38], with 15 and 21 patients with glioma recruited, respectively. Carboplatin was used as a chemotherapeutic drug, and SonoVue, an ultrasound contrast agent, was used. Magnetic resonance imaging (MRI) results revealed that the signal intensity derived from contrast agents was increased after ultrasound exposure, and carboplatin-related side effects were not observed. The patients who had no clear BBB opening had a median overall survival (OS) of 8.64 months, whereas those with a clear BBB opening had an OS of 12.94 months. These studies suggest that an implantable ultrasound device would be safe and feasible for glioma therapy.

NaviFUS utilizes multi-channel hemispherical phased-array ultrasound at 500 kHz. The Medtronic Stealth Station S7 neuronavigation system with a supplementary phantom was used to control the focal point for ultrasound exposure. Clinical studies were reported in 2020 [39] and 2021 [40], with six patients with glioma recruited. SonoVue was also used in the clinical trial, where ultrasound exposure induced leakage of the MRI contrast agent. No side effects were observed. These studies suggested that NaviFUS would be safe for BBB opening in humans.

Targeted drug delivery to tumors in other tissues

The treatment of tumors outside the central nervous system remains a challenge. In tumor tissues, neovascular formation is induced to obtain sufficient oxygen and nutrition. The neovasculature has high permeability due to the large gap between the endothelial cells compared to tissue with normal vasculature, and the lymphatic system is not well established. Therefore, large molecules including nanomedicine can extravasate into the tumor tissues and cannot be drained from tumor tissues, resulting in accumulation, which is known as enhanced permeability and retention effects [41]. However, it was recently reported that nanomedicine (i.e., doxorubicin-loaded liposomes [Doxil]) could not efficiently accumulate in tumor tissues in humans [42]. One possibility for the insufficient accumulation of Doxil in tumor tissues is low vascular permeability in humans compared with that in an animal model [43]. Therefore, a drug delivery system that can enhance vascular permeability is desirable for efficient nanomedicine delivery. The vascular permeability enhancement achieved with the combination of ultrasound and microbubbles has been applied to drug delivery to tumors. Many compounds, such as anti-cancer drugs

with low molecular weight [44–47], nucleic acids [48, 49], and nanomedicines [50–52], exhibited improved therapeutic effects in preclinical studies with use of the combination of ultrasound and microbubbles. Clinical studies were performed with gemcitabine and SonoVue using a diagnostic ultrasound device to treat pancreatic cancer [53, 54]. Ten patients were recruited, and the therapeutic protocol did not induce toxicity or clinical side effects. In addition, the median survival in patients was increased from 8.9 months to 17.6 months. The combination of ultrasound and microbubbles might be useful for improving the therapeutic effects of gemcitabine in patients with pancreatic cancer. This combination has also been used to treat many organs such as the liver [55], kidney [56, 57], and muscle [58] with drugs delivered via blood vessels. Therefore, using ultrasound and microbubble combination, which induces enhanced vascular permeability, can be applied to various diseased tissues via blood vessels to improve efficacy without the need to develop novel drugs.

Cancer immunotherapy by the combination of microbubbles and ultrasound exposure

Immunotherapy is an attractive therapy for cancer because of the induction of cancer cell-specific immune responses without serious side effects, such as those caused by chemotherapy. Initial scientific observations for the modulation of a patient's immune system to cure cancer were performed by Fehleinsen and Busch. They independently noticed significant tumor regression after erysipelas [59]. The next advanced immunotherapy was performed by Coley, who is known as the father of immunotherapy. In 1891, Coley and his collaborators attempted to utilize the immune system for the treatment of bone cancer. They injected more than 1000 cancer patients with bacteria or bacterial products (Coley's toxins). Excellent results were reported, especially in bone and soft-tissue sarcomas [60]. However, the anti-tumor mechanism could not be identified, and the focus was on the risk of infection owing to the use of highly pathogenic bacteria. Meanwhile, oncologists began focusing on radiotherapy and chemotherapy, with the above result being overlooked for decades.

In 1991, Boon et al. reported a melanoma antigen encoding (MAGE) gene that encodes an antigen recognized by cytotoxic T lymphocytes (CTLs) [61]. Later, many tumor-associated antigens were identified and studied, which led to the discovery of the human leukocyte antigen (HLA)-binding peptides recognized by T-cell receptors. In addition, it was shown that dendritic cells (DCs) play an important role as professional antigen-presenting cells for T cells to prime the anti-tumor immune response. Recently, neoantigens, which are novel antigens expressed in cancer cells

but not in normal cells, were expected to be developed as an effective therapeutic cancer vaccine. Vaccination with various tumor-associated antigens, CTL epitope peptides, or antigen-pulsed DCs was performed. However, the effect of cancer vaccines was limited because of immunosuppression in the tumor microenvironment. In cancer immunotherapy, including cancer vaccines, the cancer-immunity cycle has been proposed as an important theory to achieve sufficient therapeutic effect (Fig. 2) [62]. This cycle is composed of seven steps: antigen release from cancer cells, antigen presentation by DCs, T-cell priming/activation, T-cell migration at the tumor tissue, T-cell infiltration into tumor tissue, and recognition and killing of cancer cells by T cells. To achieve effective cancer immunotherapy, it is necessary to prime CTLs as the primary effector cells and repeatedly activate this cancer-immunity cycle [63–65]. However, poor T-cell infiltration and activity have been observed in solid tumors. In most cases, the microenvironment in solid tumors is immunologically suppressed, resulting in limited T-cell infiltration and activation. Tumor tissue with this tumor-mediated immune suppression is known as a “cold” tumor [66, 67]. Conventional cancer immunotherapies based on the activation of tumor-specific T-cell responses, such as peptide/antigen immunization and DC-based vaccines, have shown limited clinical efficacy. To disrupt this immunological suppression and change the microenvironment in tumor tissue from “cold” to “hot,” immune checkpoint blockades (e.g., anti-PD-1/anti-PD-L1 antibodies) have been utilized [68]. Thus, OS could significantly increase in some patients after immune checkpoint blockade therapy [69]. In addition, adoptive T-cell therapies, such as chimeric antigen receptor-T (CAR-T), have been developed [70, 71]. We can select an optimal strategy for attacking cancer cells using various protocols. Despite these promising immunotherapies, approximately 60–85% of patients with solid tumors cannot benefit from these [72–74]. To solve this problem, many researchers have been developing effective cancer immunotherapy to aid in therapy [75–78].

Various protocols and methods for promoting the cancer-immunity cycle have been reported, such as antigen delivery [79, 80], cytokine/chemokine injection [66, 81–85], or regulation of immune suppression. These strategies are mainly divided into two types: acceleration of the anti-tumor immune response and disrupting immune suppression. Recently, cancer therapy with external physical energies has been focused on accelerating the anti-tumor immune response. In particular, radiation therapy has been reported to effectively prime cellular immunity, which is the most important immune response as an effector in cancer immunotherapy, by inducing immunogenic cell death of tumor cells [86, 87]. However, radiation therapy is associated with a potential hazard as patients and medical professionals suffer increasing radiation exposure, which is a limitation

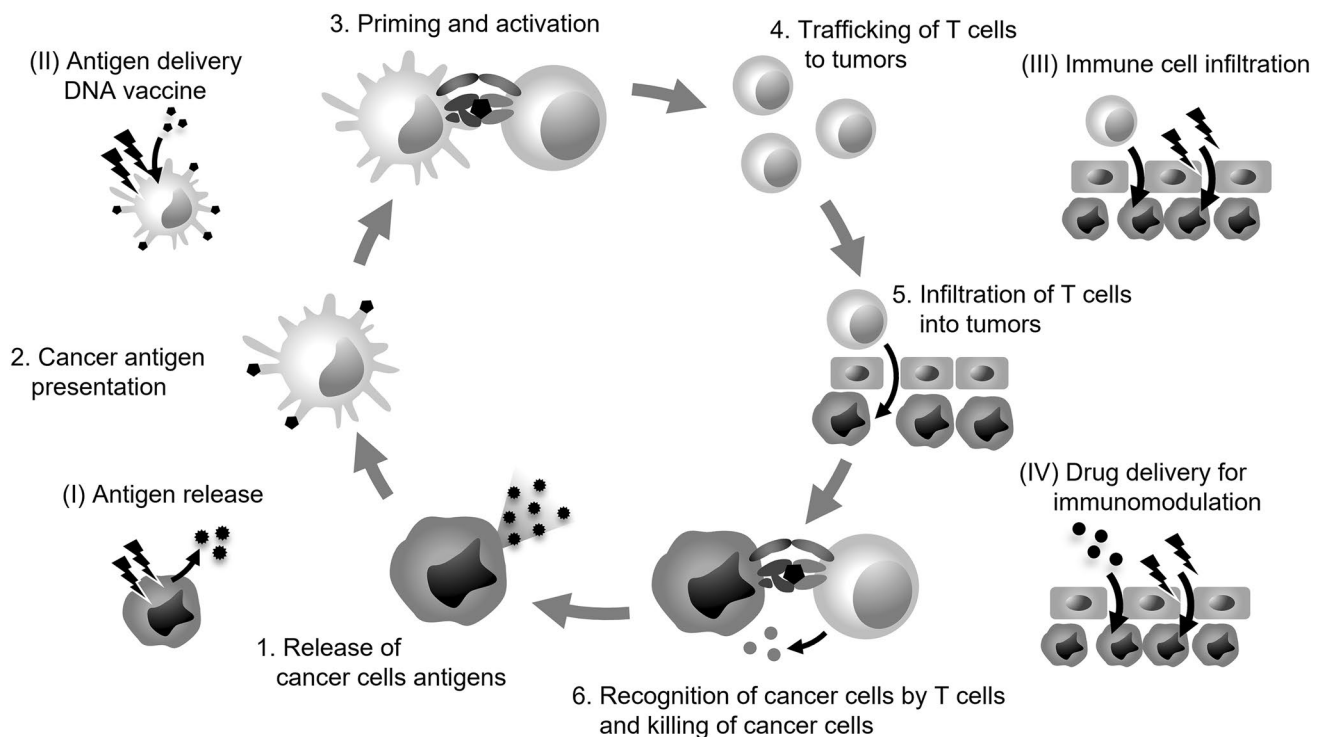


Fig. 2 Activation of cancer-immunity cycle by ultrasound and microbubbles. The generation of immunity to cancer is a cyclic process that can be self-propagating, leading to an accumulation of immune-stimulatory factors that, in principle, should amplify and broaden T-cell responses. The cycle is also characterized by inhibitory factors that lead to immune regulatory feedback mechanisms, which can halt the

development or limit immunity. This cycle can be divided into seven significant steps, beginning with the release of antigens from the cancer cells and ending with killing cancer cells. Each step is described previously herein, with the primary cell types involved and the anatomic location of the activity listed. *APCs* antigen-presenting cells, *CTLs* cytotoxic T lymphocytes

during its repetitive use. In contrast, ultrasound is external energy without the hazards of radiation and has no limitation such as that of radiation therapy. The cancer-immunity cycle can be induced by various anti-tumor immune responses, such as the delivery of antigens and cytokines, induction of immunogenic cell death, and recruitment of competent immune cells. Recently, it has been reported that therapeutic ultrasound also has the potential to induce anti-tumor immune responses [77, 88]. Moreover, the combination of ultrasound and liposomal bubbles activates cellular immunity and could be highly effective for promoting anti-tumor activities [89]. These technologies for drug delivery systems with this combination have the potential to be developed to extend effective anti-tumor immunotherapy. In this section, we introduce new challenges for priming anti-tumor immunity using ultrasound and microbubbles.

Antigen release from cancer cells

High-intensity focused ultrasound, which is used to focus ultrasonic energy on a target region from outside the body, has been utilized as an ablation therapy for cancer [88]. This therapy primarily kills cancer cells by localized heating of

the cancer tissue. Damaged cancer cells release tumor-associated antigens, danger signals such as ATP and HMG-B1, and damage-associated molecular patterns. Subsequently, DCs take up the antigens and mature by receiving maturation signals [87, 90]. Additionally, microbubbles might also enhance the mechanical and thermal effects of ultrasound exposure by inducing oscillation and cavitation. Intratumoral injection of lipid-stabilized bubbles and subsequent ultrasound exposure is reportedly effective at damaging the tumor tissue and priming cellular immunity against cancer cells [89]. The immune response suggests that the tumor-associated antigens are released from damaged cancer cells and taken up by DCs, and the antigen-derived epitope peptides lead to the presentation by MHC class I molecules. Moreover, the anti-tumor effect depends on the intensity of ultrasound exposure [89]. This suggests that the degree of damage to the tumor might be associated with the anti-tumor immune response.

Promotion of antigen presentation with professional antigen-presenting cells (APCs)

To prime antigen-specific T cells, antigens should be presented by MHC molecules on APCs, such as DCs. Priming

CD8⁺ T cells specific to cancer cells is the most crucial step after MHC class I antigen presentation on APCs. Upon vaccination via the subcutaneous or intramuscular injection of antigens, extracellular antigens are normally present on MHC class II molecules on APCs and primarily on primed CD4⁺ T cells. In this case, CD8⁺ T cells are not effectively primed. Therefore, in DC-based cancer immunotherapy, it is vital to induce the cross-presentation of extracellular antigens by DCs. Commonly, endogenous proteins are presented by MHC class I molecules. However, APCs have a cross-presentation pathway to induce MHC class I presentation even with extracellular antigens. Cross-presentation refers to the presentation of the epitope peptide derived from the extracellular antigen on MHC class I molecules in APCs. Various methods to induce the cross-presentation of tumor-associated antigens have been reported, such as antigen delivery systems with delivery carriers that can release antigens into the cytosol in DCs via endosome escape [91]. The combination of ultrasound and microbubbles has been reported as a new antigen delivery system that can directly deliver extracellular antigens into the cytosol in DCs [92, 93] by opening transient pores in the cell membranes of these cells. Extracellular antigens can then directly enter the cytosol. It is anticipated that extracellular antigens in the cytosol might be recognized as endogenous. Therefore, this antigen delivery system could be applied to antigen delivery in DC-based cancer immunotherapy [92]. In a murine melanoma lung metastasis model, this antigen delivery system was able to effectively suppress lung metastasis [93].

DNA vaccines are also an effective method for cancer immunotherapy, because they express the antigen in APC cytosol and induce MHC class I presentation. For DNA delivery, non-viral vectors are thought to be safer than viral vectors. Meanwhile, efficiency of gene delivery with non-viral vectors is not sufficient for DNA vaccines in cancer therapy. To overcome this problem, bubbles containing plasmids encoding an APC-targeted melanoma antigen have been developed [94]. Five minutes after the intravenous injection of these bubbles, the abdominal area of the mice was exposed to ultrasound. This treatment resulted in the priming of CTLs against melanoma cells and anti-tumor effects in a lung metastasis model.

Recently, cancer cell damage induced by boiling histotripsy, which mainly utilizes the mechanical effect of high-intensity focused ultrasound, was found to induce a strong immune response [95]. In this case, cancer cells were killed without inducing coagulative thermal damage releasing the tumor-associated antigens. The mechanical stress produced by boiling histotripsy promoted immunogenic cell death via the TNF-induced necrosis signaling pathway. This significantly increased the levels of damage-associated molecular patterns (HMG-B1, calreticulin, HSP70), pro-inflammatory cytokines (IFN- γ , IL-1 α , IL-1 β), and chemokines (IL-8).

Therefore, the induction of immunogenic cell death with external ultrasound energy could be a potential therapeutic approach for cancer immunotherapy.

Priming T cells and T-cell infiltration into tumor tissue

In cancer immunotherapy, it is important to evoke antigen-specific T-cell responses for a direct attack on cancer cells. To enhance T-cell response, activated T-cell therapy was developed. In early adoptive immunotherapy, CD3⁺ T cells collected from peripheral blood were activated with IL-2 and injected into the patients. In the next generation, tumor-infiltrating lymphocytes (TILs) were collected from tumor tissue and injected into the patient after TIL activation *in vitro*. However, the anti-tumor effect was limited owing to the activation of lymphocytes, which had been suppressed in the patients. Following a transition in the preparation of antigen-specific T cells, recent technologies have shown that CAR-T cells, which are artificial T cells, specifically bind and kill antigen-expressing cells similar to CTLs. CAR-T cell immunotherapy is an adoptive immunotherapy for cancer and has been approved for B-cell lymphomas. The approval of CAR-T cell immunotherapy marks the beginning of a new generation of immunotherapy that uses a cell as a drug. CAR-T cells with synthetic receptors are designed based on the principles of T-cell receptors and costimulatory signaling, enabling the non-MHC-restricted targeting of tumors. The target of CAR-T cell therapy is mainly blood cancers, such as leukemia and lymphoma. Unfortunately, CAR-T cell therapy is rarely applied to solid tumors because of the low infiltration of CAR-T cells into the tumor tissue [96]. To improve the efficiency of infiltration, some experiments have used chemokines to enhance the expression of adhesion molecules for T-cell attachment on endothelial cells in tumor tissue [78]. In contrast, the combination of ultrasound and microbubbles could directly affect endothelial cells via the oscillation and cavitation of microbubbles [97]. The combination of ultrasound and microbubbles can enhance natural killer (NK) cell infiltration into tumor tissue to develop an effective NK-cell-based adoptive immunotherapy [98–101]. NK cells are cytotoxic lymphocytes of the innate immune system that exhibit anti-tumor activity. Recently, it was reported that the loss of MHC class I expression on tumor cells led to the avoidance of T-cell recognition via immune escape [102]. The promotion of NK-cell delivery into the tumor tissue contributes to the anti-tumor effect. Alkins et al. reported that HER2-specific NK-92 cells accumulated in breast cancer cells implanted in the brain of mice in a metastatic breast cancer model by opening the BBB with ultrasound and microbubble combination [99]. Yang et al. reported that ultrasound and microbubble combination could improve NK-92MI cell infiltration into ovarian tumor

tissue [98]. They also showed that CXCL1 might be involved in the process of NK-cell infiltration. The combination of ultrasound and microbubbles can open the tight junctions between endothelial cells and induce chemokine secretion to facilitate the infiltration of lymphocytes [97]. The infiltrated lymphocytes would improve the microenvironment in tumor tissue for the activation of the anti-tumor immune response.

Recognition and killing of cancer cells by T cells

CTLs recognize tumor cells via T-cell receptors and cause their death. In this step, when the microenvironment of tumor tissue is under immune suppression, reduction of MHC class I molecule expression on tumor cells, suppression of CD8⁺ T cells via PD-1/PD-L1 signaling, and an increase in regulatory T lymphocytes and Th1/Th2 imbalances are observed. To improve this environment, various approaches, such as the induction of innate immune responses, cytokine therapy, and immune checkpoint blockades (ICBs), have been developed. In particular, ICBs have cured patients with stage IV cancer. However, the therapeutic effects are limited because of the immunosuppressive tumor environment (“cold” tumor) [66]. IL-12 is an important cytokine for changing the Th1/Th2 balance to Th1 dominant, which results in the induction of an immunoreactive tumor microenvironment (“hot” tumor) [66]. During cytokine therapy, frequent injection of IL-12 is needed because of the short half-life of IL-12, inducing systemic toxicity. IL-12 gene therapy is attractive for prolonging gene expression compared to that with cytokine injection. For IL-12 gene therapy, the combination of lipid-stabilized bubbles and ultrasound exposure was examined to develop the local delivery of IL-12-encoded plasmid DNA as a non-viral vector system [103]. A significant anti-tumor effect was observed with this gene delivery method. This anti-tumor effect in repetitive gene delivery was compared with IL-12 gene therapy using an adenoviral vector system with a single injection [104].

The hypoxia-inducible factor (HIF-1) and vascular endothelial growth factor (VEGF) pathways contribute to immune suppression in the tumor microenvironment. Therapy with anti-VEGF and anti-VEGF receptor antibodies has been performed. In this therapy, tumor vascular normalization is anticipated to be an essential part of antiangiogenic therapy [105]. A new strategy was developed to target vascular tumors with clots and poor blood flow in this therapy. Li et al. reported that the mechanical effects of the oscillation and cavitation of microbubbles owing to ultrasound could promote tumor perfusion, resulting in enhanced T-cell infiltration and delivery of the anti-PD-L1 antibody into tumor tissue [106]. Tumor vascular normalization with ultrasound and microbubbles can be induced under a lower ultrasound intensity (mechanical index: 0.4) similar to that of ultrasound

imaging. Therefore, tumor vascular normalization in this system can be performed using real-time ultrasound imaging of vascular flow. This therapeutic system can be combined with diagnostics to check blood flow normalization. In summary, this system comprises “theranostics”—a new concept of therapeutics and diagnostics based on the combination of ultrasound and microbubbles [107]. Soon, we will be able to reap the benefits of theranostics based on ultrasound and microbubble technologies for cancer immunotherapy.

Immunomodulation of the tumor microenvironment

It was recently reported that the use of ultrasound and microbubbles for tumor tissue treatment could change the tumor microenvironment. Hunt et al. reported that the infiltration of CD45⁺/CD3⁺ T cells was increased in tumor tissue treated with ultrasound and microbubbles [108]. Suzuki et al. also reported that CD8⁺ T cells significantly infiltrated into tumor tissue after treatment with ultrasound and liposomal bubbles and played a vital role in suppressing tumor growth [103]. This shows that the combination of ultrasound and microbubbles would be a novel method of immunomodulation to change the tumor microenvironment from “cold” to “hot” in a noninvasive and effective manner. At multiple steps in the cancer-immunity cycle, the combination of ultrasound and microbubbles promotes the anti-tumor immune response. Therefore, this strategy has potential as cancer immunotherapy based on immunomodulation of the microenvironment in tumor tissue.

Future prospects

The combination of ultrasound and microbubbles is a promising strategy for tumor-targeted drug delivery via enhanced vascular permeability with noninvasiveness and selectivity. In fact, there are several ongoing clinical studies of this strategy for tumors of the brain and other organs. The combination of focused ultrasound and microbubbles with imaging guidance can restrict the enhanced vascular permeability, making it suitable for primary cancer therapy. However, some improvements in ultrasound are needed for multiple metastases and large tumor therapy. A three-dimensional probe would be one approach for homogenous ultrasound exposure to tumors. We have previously reported that three-dimensional probes can be used for drug delivery to tissues [57]. In addition, microbubbles play an important role in the induction of mechanical effects, and the development of microbubbles is required to achieve efficient and safe drug delivery. Most microbubbles are composed of inner gas and outer shells. These components affect the stability and efficiency of drug delivery [109–112]. The fabrication

of microbubbles with homogenous sizes and the incorporation of targeting ligands would also improve the efficiency of drug delivery [113, 114]. The mechanical and thermal effects induced by the combination of ultrasound and microbubbles are also attractive for cancer immunotherapy. This combination can deliver cytokines and enhance the infiltration of immunocompetent cells into tumor tissues, resulting in efficient cancer therapy. In addition, ultrasound and microbubbles together can induce the release of antigens and danger signals from cancer cells by enhancing cell membrane permeability and damage to tumor cells. Although the detailed mechanism of enhanced anti-cancer immune responses induced by the combination of ultrasound and microbubbles has not been clarified, this technique could contribute to multiple steps of the cancer-immunity cycle to prime effective anti-tumor immunity. When using the combination of immunotherapy with ultrasound and microbubbles, appropriate enhanced cellular permeability and damage to tumor cells are important, because excess damage might induce an immunosuppressive action.

Conclusion

The combination of ultrasound and microbubbles can induce many biological reactions via mechanical and thermal effects. Enhanced vascular permeability is a valuable approach for drug delivery to tissues and can improve therapeutic effects on diseases. Although we focused on tumors in this review, other organs and diseases have also been treated with ultrasound and microbubbles. Furthermore, immunomodulation via ultrasound and microbubbles has recently received much attention. Ultrasound and microbubbles together activate the immune system via mechanical or thermal effects. Immunomodulation would be a breakthrough to improve the problem of conventional therapies such as ICBs and CAR-T cell therapy. To achieve efficient drug delivery and immunomodulation, the development and optimization of an ultrasound application system with microbubbles for organs or diseases is essential. In addition, although we did not describe other applications of ultrasound and microbubbles, combinatorial treatment has been applied to imaging or ablation therapy. Ultrasound and microbubbles have great potential to be established as therapeutic systems. In conclusion, although advanced studies are still required, the combination of ultrasound and microbubbles can be applied to many therapies to ensure drug delivery and immunomodulation.

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Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

Ethical approval This article does not contain any studies with human or animal subjects performed by any of the authors.

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